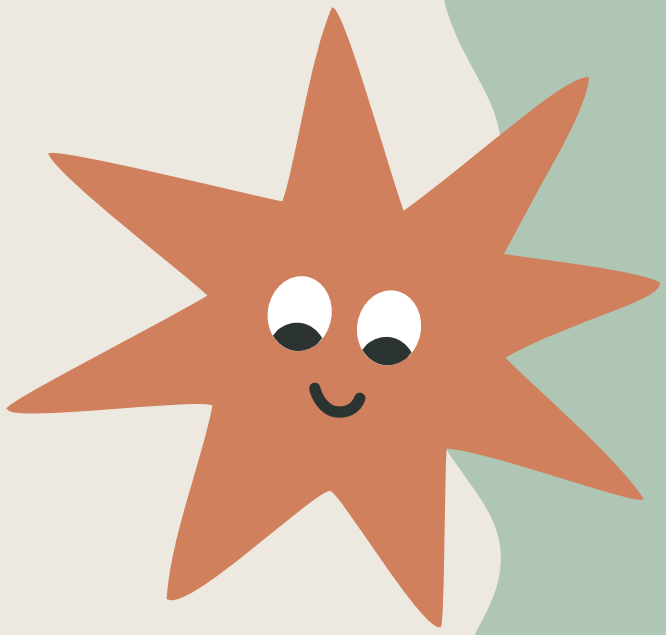
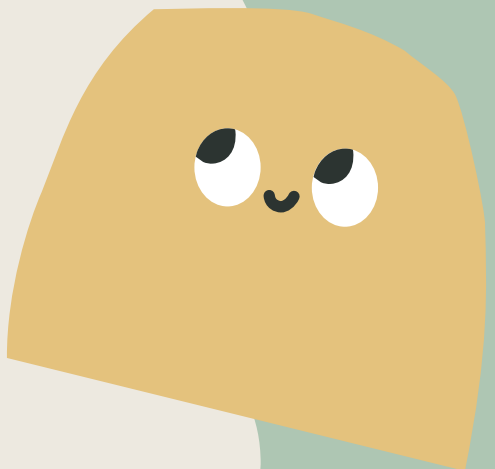
Informed and Uninformed Search

Lecture 9





Uninformed Search

- Also called **blind search**
- It uses only the information provided by the search problem
- It does **not** know which direction is closer to the goal
- It uses:
  - Initial state
  - Actions
  - Transition model
  - Goal test
  - Path cost

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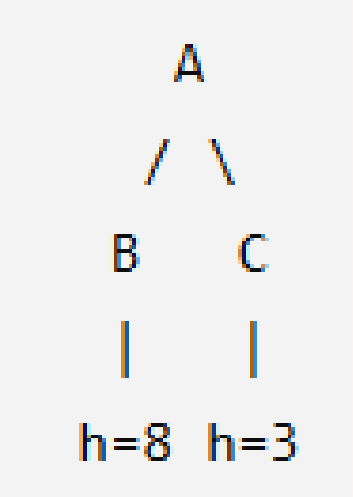
Informed Search

- **Informed search** uses additional knowledge about the problem to decide which state is more promising
- This additional knowledge is called a **heuristic**
- Heuristic = an estimate that helps us decide which path looks more promising
- Suppose an AI needs to travel from **Lahore to Islamabad**
- If it knows the approximate distance of each city from Islamabad, it can prioritize cities that appear closer to the destination

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Heuristic Function

- The notation for heuristic function is: **h(n)**
- Where:
  - **n** = current node/state
  - **h(n)** = estimated cost from node **n** to the goal



- Example
- If the goal is somewhere beyond B and C, the search may prefer **C** because h(C) = 3 is smaller

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Uninformed vs Informed Search

Feature	Uninformed	Informed
Other name	Blind Search	Heuristic Search
Uses heuristic?	❌ No	✅ Yes
Goal direction knowledge	No	Yes
Guidance	Limited	More guided
Usually explores	More states	Fewer promising states
Example	BFS, DFS	Greedy, A*

Simple Real-Life Example

- Imagine searching for a classroom in a university
- Uninformed Search
  - You check rooms one by one without knowing where the classroom is
- Informed Search
  - You have a map showing that the classroom is probably in **Block B**, so you search there first
  - Map information = heuristic knowledge

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Uninformed Search

- Advantages
  - Simple
  - Easy to implement
  - Does not require domain-specific knowledge
- Disadvantages
  - May explore many unnecessary states
  - Can be slow
  - Can consume significant memory

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Informed Search

- Advantages
  - More directed towards the goal
  - Can reduce unnecessary exploration
  - Often more efficient
- Disadvantages
  - Requires heuristic information
  - Quality depends on heuristic
  - Designing a good heuristic can be difficult

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Search Algorithms Types

- Uninformed search algorithms
  - Breadth-First Search – BFS
  - Depth-First Search – DFS
  - Iterative Deepening Search
- Informed search algorithms
  - A* Search
  - Greedy Best-First Search (Greedy Search)

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